A piecewise quadratic function without a polynomial lattice representation

Abstract

We construct a continuous function on $\mathbb{R}^{72}$ with a finite closed semialgebraic cover on which its labels are homogeneous quadratic polynomials, and prove that it has no finite lattice expression in real polynomials of arbitrary degrees. Positive homogeneity imposes a necessary finite condition on signs of linear and quadratic polynomials. Pairs of matrices with opposite determinant signs violate that condition. The construction concerns the unrestricted Pierce–Birkhoff assertion; its displayed partition uses a determinant of degree six.

A finite polynomial lattice expression is obtained from finitely many real polynomials by finitely many applications of maximum and minimum. By distributivity, such expressions are precisely finite maxima of finite minima of polynomials. A continuous function is piecewise polynomial here if a finite collection of closed semialgebraic sets covers its whole domain and the function agrees with a polynomial on each set.

Identify $M_6(\mathbb{R}) \times M_6(\mathbb{R})$ with $\mathbb{R}^{72}$, using independent matrix coordinates X, Y. For $1 \le i \le 6$ and signs $\sigma = (\sigma_j)_{j \ne i} \in \{-1, 1\}^5$, put

$$q_{i,\sigma}(X,Y) = (XY)_{ii} - \sum_{j \ne i} \sigma_j (XY)_{ij}, \qquad h(X,Y) = \min_{i,\sigma} q_{i,\sigma}(X,Y).$$

There are 192 homogeneous quadratic forms $q_{i,\sigma}$, all with integer coefficients. Define

$$f(X,Y) = \begin{cases} h(X,Y), & h(X,Y) > 0 \text{ and } \det X > 0, \\ 0, & \text{otherwise.} \end{cases} \tag{1}$$

Theorem 1. *The function $f : \mathbb{R}^{72} \to \mathbb{R}$ in (1) is continuous and has a finite closed semialgebraic polynomial cover. It has no finite lattice expression in real polynomials, with no restriction on their degrees.*

1 The whole-space polynomial cover

Writing $P = XY$, minimization over the row signs gives

$$h(X,Y) = \min_i \left(P_{ii} - \sum_{j \ne i} |P_{ij}| \right).$$

If $h > 0$, then P has positive diagonal and is strictly diagonally dominant by rows. To see that it is invertible, suppose $Pv = 0$ with $v \ne 0$, and choose i maximizing $|v_i| > 0$. The row equation gives

$$P_{ii}|v_i| \le \sum_{j \ne i} |P_{ij}|\,|v_j| \le \left(\sum_{j \ne i} |P_{ij}| \right) |v_i| < P_{ii}|v_i|,$$

a contradiction. In particular, $\det X \neq 0$ on $\{h > 0\}$.

On that open set the sign of $\det X$ is locally constant, so f is continuous there. It is locally zero on $\{h < 0\}$. At $h = 0$, its value is zero and

$$0 \leq f \leq \max(h, 0),$$

which proves continuity at all remaining points, including singular X and Y.

Enumerate the quadratics as $q_1, \ldots, q_{192}$, and define

$$F_0 = \{\det X \leq 0\} \ \cup \ \bigcup_j \{q_j \leq 0\},$$

$$F_i = \{\det X \geq 0,\ q_j \geq 0 \text{ for all } j,\ q_i \leq q_j \text{ for all } j\}, \qquad 1 \leq i \leq 192.$$

These 193 closed semialgebraic sets cover the whole space. Use label 0 on F_0 and label q_i on F_i. On F_i, the minimum $h = q_i$ is nonnegative. If it is positive, invertibility and $\det X \geq 0$ give $\det X > 0$, so the label equals f. On $F_i \cap F_0$, either a q_j is zero or $\det X = 0$; in the latter case strict positivity of all q_j is impossible. Thus $q_i = 0$ on this overlap. On $F_i \cap F_j$, both labels equal the common minimum. This verifies the asserted polynomial cover.

For every $t > 0$, simultaneous scaling gives

$$f(tX, tY) = t^2 f(X, Y). \tag{2}$$

Indeed, each $q_{i,\sigma}$ scales by t^2, whereas $\det(tX) = t^6 \det X$.

2 A necessary condition from positive homogeneity

Lemma 1. *Let $g : \mathbb{R}^N \to \mathbb{R}$ satisfy $g(tz) = t^2 g(z)$ for every $t > 0$. If g has a finite lattice expression in arbitrary real polynomials, there is a finite family of polynomials of degree at most two such that any two points sharing all signs in that family have the same sign of g.*

Proof. Write $g = \mathcal{L}(p_1, \ldots, p_m)$, where $\mathcal{L}$ is a fixed finite expression using only maximum and minimum. Decompose each leaf into its homogeneous parts:

$$p_j = c_j + \ell_j + Q_j + \sum_{d \geq 3} P_{j,d}.$$

Here ℓ_j has degree one and Q_j degree two; either can be zero. Let $\mathcal{T} = \{\ell_j, Q_j : 1 \leq j \leq m\}$.

For a fixed point z, positive scaling commutes with the order operations, and hence for every $t > 0$,

$$g(z) = \mathcal{L}\left(\frac{p_1(tz)}{t^2}, \ldots, \frac{p_m(tz)}{t^2}\right).$$

Each entry has an extended-real limit $u_j(z)$ as $t \downarrow 0$:

$$u_j(z) = \begin{cases} \operatorname{sgn}(c_j)\,\infty, & c_j \neq 0, \\ \operatorname{sgn}(\ell_j(z))\,\infty, & c_j = 0,\ \ell_j(z) \neq 0, \\ Q_j(z), & c_j = 0,\ \ell_j(z) = 0. \end{cases}$$

Terms of degree at least three tend to zero after division by t^2 at each fixed z. Finite minimum and maximum are continuous on the extended real line. For example, conjugation by the increasing homeomorphism $(2/\pi)\arctan : [-\infty, \infty] \to [-1, 1]$ reduces this to their ordinary continuity. No addition of infinities occurs. Therefore

$$g(z) = \mathcal{L}(u_1(z), \ldots, u_m(z)). \tag{3}$$

Extend sign by $\operatorname{sgn}(-\infty) = -1$ and $\operatorname{sgn}(+\infty) = 1$. Every nondecreasing map between totally ordered sets preserves minimum and maximum. Applying sign to the exact equality (3) gives

$$\operatorname{sgn} g(z) = \mathcal{L}(\operatorname{sgn} u_1(z), \ldots, \operatorname{sgn} u_m(z)).$$

The inputs on the right are determined by the fixed constants c_j and the signs of $\ell_j(z), Q_j(z)$. Thus equal joint signs for $\mathcal{T}$ imply equal signs of g.

The limit is taken before sign is applied; sign is not interchanged with a limit at zero. Only pointwise convergence is used. □

3 All quadratic equations on an orbit

Put

$$D = \operatorname{diag}(1, 1, 0, 0, 0, 0), \qquad E = \operatorname{diag}(0, 0, 1, 1, 0, 0),$$

and consider

$$\mathcal{O} = \{(UDV^{-1}, VEU^{-1}) : U, V \in \mathrm{GL}_6(\mathbb{R})\}.$$

Its pairs have ranks two and two, and satisfy $XY = YX = 0$.

Lemma 2. *A real polynomial of total degree at most two vanishes on $\mathcal{O}$ if and only if it is a constant real linear combination of the entries of XY and YX.*

Proof. The reverse implication is immediate. For nonzero real a, b, replace U by

$$U \operatorname{diag}(a, a, b^{-1}, b^{-1}, 1, 1)$$

and keep V fixed. This replaces (X, Y) by (aX, bY). Consequently, separating coefficients in a, b, each bihomogeneous component of a polynomial vanishing on $\mathcal{O}$ vanishes separately.

The X-projection and the Y-projection of $\mathcal{O}$ are both the entire rank-two matrix locus, by independent changes of basis on the left and right. Its closure contains every rank-one matrix and zero. A homogeneous linear polynomial vanishing there is zero by evaluation at matrix units. For a homogeneous quadratic, these evaluations eliminate the squared-coordinate coefficients, and evaluation at sums of two distinct matrix units eliminates all mixed coefficients. Each such sum has rank at most two. The constant component is also zero. Thus only a bilinear polynomial $B(X, Y)$ can remain.

We next show that every nonzero rank-one pair

$$X = uv^T, \qquad Y = st^T, \qquad t^T u = 0, \quad v^T s = 0 \tag{4}$$

lies in the Euclidean closure of $\mathcal{O}$. Regard X as a map from a second vector space to a first, and Y in the reverse direction. In the first space choose a basis with first vector u, a second vector on which t^T is one, and all other vectors in $\ker(t^T)$. In the second space choose a first vector on which v^T is one, second vector s, and all other vectors in $\ker(v^T)$. The two zero-pairing conditions make these choices possible. In these bases $X = E_{11}$ and $Y = E_{22}$, where E_{ij} are matrix units.

Let P interchange coordinates two and three. For $\delta \neq 0$, set

$$U_\delta = P\,\mathrm{diag}(1, \delta, 1, \delta^{-1}, 1, 1), \qquad V_\delta = P.$$

Directly,

$$U_\delta D V_\delta^{-1} = E_{11} + \delta E_{33}, \qquad V_\delta E U_\delta^{-1} = E_{22} + \delta E_{44}.$$

These tend to the normalized rank-one pair. Composing with the two fixed changes of basis proves the closure assertion. Hence B vanishes on all pairs (4); if one matrix is zero it vanishes by bilinearity.

Write

$$B(X, Y) = \sum_{i,j,k,l} b_{ij,kl} X_{ij} Y_{kl}, \qquad T(A, C) = \sum_{i,j,k,l} b_{ij,kl} A_{il} C_{kj}.$$

The second expression defines a bilinear form on two matrix spaces, and

$$B(uv^T, st^T) = T(ut^T, sv^T). \tag{5}$$

The two arguments on the right range independently over all rank-one traceless matrices: their trace conditions are exactly $t^T u = 0$ and $v^T s = 0$. Such matrices span the entire traceless matrix space. They include all off-diagonal units, and

$$E_{ii} - E_{jj} = (e_i + e_j)(e_i - e_j)^T + E_{ij} - E_{ji}$$

supplies all diagonal differences. Bilinearity therefore shows that T vanishes on the product of the two full traceless spaces.

Decomposing each argument into a traceless matrix plus its trace times $I/6$ now gives

$$T(A, C) = \mathrm{tr}(A)L(C) + \mathrm{tr}(C)M(A),$$

where one can take

$$L(C) = \frac{T(I, C)}{6} - \frac{\mathrm{tr}(C)T(I, I)}{36}, \qquad M(A) = \frac{T(A, I)}{6}.$$

Substituting in (5), the first summand is $L(YX)$, and the second is $M(XY)$. The trace formula holds for all A, C, so this gives $B(X, Y) = L(YX) + M(XY)$ for arbitrary rank-one X, Y, without requiring their products to vanish. Both sides are bilinear and rank-one matrices span the full matrix space. The identity therefore holds for all X, Y, as required. $\square$

4 Pairs with identical signs for any finite quadratic family

Lemma 3. *For any finite family of real polynomials $p_1, \ldots, p_M$ of degree at most two, there are two actual real matrix pairs z_+, z_- with identical signs for every p_j, but $f(z_+) > 0$ and $f(z_-) = 0$.*

Proof. For each p_j not identically zero on $\mathcal{O}$, the function

$$(U, V) \longmapsto p_j(UDV^{-1}, VEU^{-1})$$

is a nonzero rational function of the entries of U, V. Multiplication by $(\det U \det V)^2$ clears its denominators and gives a nonzero polynomial numerator. The product of these finitely many nonzero numerators is nonzero; an empty product is one.

A nonzero real polynomial cannot vanish on a nonempty Euclidean open set. Indeed, induction on the number of variables reduces vanishing on an open box to the univariate root bound by comparing coefficient polynomials. Choose an open neighborhood of (I, I) where both determinants are positive. There are U, V in this neighborhood at which all the indicated numerators are nonzero. Thus every p_j not vanishing on $\mathcal{O}$ is nonzero at the common point

$$o = (UDV^{-1}, VEU^{-1}).$$

Fix these U, V. For $\epsilon > 0$ and $\eta \in \{-1, 1\}$, put

$$D_X^\eta = \operatorname{diag}(1, 1, \epsilon^2, \epsilon^2, \epsilon, \eta\epsilon), \qquad D_Y^\eta = \operatorname{diag}(\epsilon^2, \epsilon^2, 1, 1, \epsilon, \eta\epsilon),$$

$$X_\eta = UD_X^\eta V^{-1}, \qquad Y_\eta = VD_Y^\eta U^{-1}.$$

Both pairs tend to o as $\epsilon \downarrow 0$. Exact multiplication and determinants give

$$X_\eta Y_\eta = Y_\eta X_\eta = \epsilon^2 I, \qquad \det X_\eta = \frac{\det U}{\det V}\,\eta\epsilon^6. \tag{6}$$

If p_j vanishes on $\mathcal{O}$, Lemma 2 and (6) give exactly equal values on both pairs. If $p_j(o) \neq 0$, continuity gives the same strict sign on both pairs for all sufficiently small positive ϵ. There are finitely many such requirements, so a single positive real ϵ makes every sign agree.

All 192 forms $q_{i,\sigma}$ equal ϵ^2 on both pairs. Since $\det U / \det V > 0$, their X determinants have opposite signs. Therefore

$$f(X_+, Y_+) = \epsilon^2 > 0, \qquad f(X_-, Y_-) = 0.$$

The choices are made in the order: the finite polynomial family, then U, V, then ϵ. The function f is fixed throughout. □

5 Conclusion

Proof of Theorem 1. Section 1 verifies continuity and the finite closed polynomial cover on all of $\mathbb{R}^{72}$. If f had any finite polynomial lattice expression, positive homogeneity (2) and Lemma 1 would make its sign determined by some finite family of degree-at-most-two polynomials. Lemma 3 supplies two real points with identical signs for that family and different signs of f, a contradiction. □

The displayed semialgebraic cover uses a determinant of degree six. The theorem does not assert that f has one shared polynomial label on each entire sign realization of some finite family of quadratic partition polynomials. That additional partition requirement is different from the unrestricted whole-space assertion disproved here.